

ODON INEZA

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EDUCATION

University of Notre Dame

Bachelor of Science in Computer Science; Mathematics Minor

Notre Dame, IN

GPA: 3.73 / 4.00

EXPERIENCE

University of Notre Dame

June 2026 – August 2026

Research Intern: Face Image Quality CNN

Notre Dame, IN

- Designed and trained a compact SmallResNet (0.33M params) in PyTorch to predict OFIQ face-image quality on 70K FFHQ images, achieving MAE 1.32, Pearson $r = 0.866$, and Spearman $\rho = 0.867$ on a 7K held-out set ($2\times$ better than a mean baseline) while cutting overfitting from a $27\times$ to $1.1\times$ val/train MSE ratio vs. ResNet-18
- Built the model as a frozen differentiable quality critic for a GAN generator loss, enabling millisecond OFIQ-like feedback during generative training instead of multi-second full OFIQ scoring
- Delivered a reproducible training/evaluation pipeline plus Streamlit/Flask demos

Pivot Access Ltd

Aug 2023 – Jul 2025

Full Stack Developer; Front End Developer & Research Intern

Kigali, Rwanda

- Maintained production infrastructure with Kubernetes, Docker, Prometheus, Grafana, Loki, and New Relic, improving observability across multi-service deployments
- Automated CI/CD and monitoring workflows, reducing manual release steps and shortening incident detection time for live services
- Shipped responsive React + Tailwind CSS interfaces that improved usability and engagement across client-facing web applications
- Analyzed multivariate product data in PostgreSQL and visualized results with Matplotlib to drive research-backed UI and feature decisions

PROJECTS

Brain Tumor MRI Classification | Python, TensorFlow/Keras, Streamlit, scikit-learn

Spring 2026

- Built an end-to-end ML pipeline classifying 7,200 MRI slices into 4 classes (glioma, meningioma, pituitary, no tumor); compared Logistic Regression, scratch CNN, modified CNN, and MobileNetV2
- Achieved 91.4% validation and 86.7% test accuracy with MobileNetV2 transfer learning (vs. 25% majority-class baseline); reduced CNN generalization gap from 9.8% to 2.8% via per-sample normalization
- Deployed a Streamlit dashboard for side-by-side inference across all 4 models on uploaded MRI slices

NanoChat (Tiny GPT Playground) | PyTorch, TypeScript, React, Vite

Spring 2026

- Implemented a trainable GPT-style language model from first principles, covering tokenization, embeddings, attention, and transformer blocks
- Trained models in PyTorch and exported weights for fully in-browser TypeScript inference with a React playground for prompting, sampling, and inspecting generation with no API dependency

NYC Taxi Analytics Dashboards | Python, Plotly, Jupyter, HTML/JS

Fall 2025

- Analyzed NYC TLC yellow-cab trip data and built 12 interactive Plotly visualizations covering temporal demand, fares/tips, congestion proxies, geography, and fleet utilization
- Automated export of notebook charts to a static multi-page site (gallery + 12 viz pages) for reproducible, shareable data storytelling

Brilliant Sciences Learning Platform | Next.js, React, Firebase, Redux, Tailwind

Fall 2025

- Built a full-stack science learning platform with course uploads, video classrooms, quizzes, progress tracking, grading, and instructor admin workflows

TECHNICAL SKILLS

Languages: Python, Java, C/C++, SQL (PostgreSQL), JavaScript, TypeScript, HTML/CSS

ML / Data: PyTorch, TensorFlow/Keras, scikit-learn, pandas, NumPy, Matplotlib, Plotly, Streamlit

Frameworks: React, Next.js, Node.js, Flask, FastAPI, Spring Boot, Firebase, Redux

Tools: Git, Docker, Kubernetes, Prometheus, Grafana, Loki, New Relic, Vercel, VS Code, Jupyter